ChatGPT

App-Ready Knowledge Base for Music Catalog Ingestion, Reconciliation, Analytics, and Valuation

Executive summary

A royalty administration app needs a **knowledge base (KB)** that is simultaneously (a) **machine-ingestible**, (b) **audit-defensible**, and (c) **rights-aware**. The core design principle is to treat every incoming statement as evidence that must be traceable through three layers:

1) **Raw ingest (immutable)**: the vendor's original rows/pages/files.
2) **Canonical ledger (normalized)**: a unified schema that preserves provenance, identifiers, periodization, currency, and sign conventions.
3) **Derived analytics & valuation (reproducible)**: decay fits, concentration, scenario forecasts, and DCF outputs that can be rebuilt from the ledger.

You should calibrate your internal requirements to the most explicit statutory/regulated transparency regime you can find—US blanket mechanical streaming as administered by **The Mechanical Licensing Collective** [1], where regulations require clear reporting and, for adjustments, identification of the **original reporting period** being adjusted. [2]

The KB should be "standards-forward":

- Composition identifiers: **ISWC** (works) and **IPI** (parties) are core to society documentation and royalty allocation workflows. [3]
- Recording identifiers: **ISRC** is the stable key for sound recordings and is foundational for master-side ingestion and SoundExchange reporting. [4]
- Name identity: **ISNI** is a widely used ISO identifier to disambiguate creators and organizations in metadata and rights contexts. [5]
- Data exchange: DDEX's **DSR** (sales/usage reporting) and **MWDR** (work rights communication) provide a practical vocabulary for what fields exist and how they are expected to travel. [6]

Finally, your KB must separate **earned** vs **cash** realities. Advances, recoupment, reserves, and retro adjustments should not corrupt your "earned decay" series; they belong in separate ledgers and then flow into cash forecasting only as configured.

The worked examples below labeled **"Computed from uploaded statements"** are derived directly from your uploaded Warner/"Tango" statement CSV + PDF set and illustrate the system design in practice.

Canonical data model and schemas

A robust royalty app is easier to scale when you standardize around **a small set of first-class entities** and enforce canonical IDs everywhere.

Core entities and why they exist

Catalog
Defines the commercial boundary: owner, rights scope, start/end, included assets.

Party
Rightsholders, administrators, payees, DSPs, societies, labels, publishers. Store multiple identifiers (IPI, ISNI, vendor IDs). IPI is explicitly designed for documentation/distribution/accounting processes across societies. [7]

Work (Composition)
Keyed by internal work ID plus ISWC when known. ISWC is the global identifier for musical works used by value-chain participants. [8]

Recording (Sound Recording / Master)
Keyed by internal recording ID plus ISRC when known. ISRC is designed to uniquely identify one recording version and persist across markets. [9]

RightsClaim / Share
A time-valid, territory-valid claim to a percentage of a right (mechanical, performance, sync, etc.). DDEX MWDR guidance distinguishes share types and expectations around communicating them consistently. [10]

Statement
A document-level object: issuer, period, currency, version, settlement lag.

StatementLine
The atomic unit: the row-level royalty event (or summarized line) you can aggregate into statements and time series.

AccountTransaction
Non-line-item flows: advances, recoupment, payments, charges, retro adjustments that may not be part of line-level royalties but materially affect cash and balances.

FXRate
A reproducible currency conversion record with policy metadata. DDEX explicitly models transaction currency vs reporting currency vs invoicing currency as distinct concepts. [11]

Crosswalk / Mapping
Many-to-many mapping between internal IDs, ISWC/ISRC, and provider IDs, with confidence and override provenance.

Dispute / AuditCase
A structured lifecycle that can point back to evidence and ledger deltas. Audit rights and audit programs exist explicitly in the MLC ecosystem, which is a useful benchmark for governance design. [12]

Canonical schema patterns

Below is the **design pattern** the KB should encode:

- Every entity has a stable `uid` (string) and an optional list of `external_ids`.
- Every monetary amount records `amount`, `currency`, and `fx_policy_ref` (even if base currency uses 1.0 FX).
- Every time-based record has:
- `statement_period` (when reported),
- `activity_period` (when earned/used),
- `cash_posting_date` (when booked/paid).

Canonical JSON examples

The KB should ship with example records to test parsers and downstream logic. (In the final section, you'll get a delivery-ready JSON + YAML "machine pack" that includes these.)

Key canonical fields for statements should support regimes like MLC where statements can be delivered via portal access and must preserve adjustment provenance. [13]

Also, consider adopting database-oriented field naming for interoperability. The US Copyright Office's rules for the MLC musical works database explicitly contemplate including ISWC, ISRC, ISNI/IPI, and data provenance for sound recording information—this is an excellent blueprint for your own canonical fields. [14]

ETL and cleaning rules

ETL flowchart

```
flowchart TD
  A[Ingest Raw Files\nPDF/CSV/XLSX/TSV/API] --> B[Raw Staging Store\nimmutable + checksum]
  B --> C[Parse Layer\nstatement parser per provider]
  C --> D[Canonical Normalize\nschema + types + sign rules]
  D --> E[Identifier Mapping\nISWC/ISRC/IPI/ISNI + provider IDs]
  E --> F[Periodization Engine\nactivity vs statement vs cash]
  F --> G[Classification Engine\nrights type + channel + flags]
  G --> H[Ledger Tables\nStatementLine + AccountTransaction]
  H --> I[Reconciliation Controls\ncontrol totals + balance checks]
  I --> J[Analytics + Valuation\nreproducible models + snapshots]
```

Cleaning rules (field-level)

Numeric normalization - Coerce numeric strings; enforce decimals; persist original/raw values separately. - Enforce a single sign convention: - royalties earned/payable are positive, - reversals, chargebacks, withholding tax are negative (but explicitly flagged).

String normalization - Titles: uppercase, trim, collapse whitespace, strip punctuation, normalize “feat/featuring”. - Parties: normalize “Last, First” vs “First Last”, remove punctuation, expand common legal suffixes (LLC/Inc.).

Territory normalization Your uploaded statement data contains inconsistent territory labeling (e.g., “US”, “United States”, “UNITED STATES”). The KB should enforce canonical territory codes (ISO 3166-1 alpha-2 is the simplest) and store the raw name as provenance.

A pragmatic approach: - `territory_raw` = original string - `territory_iso2` = mapped code - `territory_confidence` = {exact, dictionary, fuzzy, manual}

Periodization rules

Define three time axes and never mix them unintentionally:

1) **Activity period**: when the usage/earning occurred.
2) **Statement period**: the reporting window covered by the statement.
3) **Cash period**: when funds were posted/paid.

For regulations, note that adjustment reporting should retain original provenance; for example, MLC adjustment reporting requires clearly indicating the original reporting period being adjusted. That principle should be generalized to your internal model. [15]

Allocating multi-month ranges

Rules your KB should encode:

- If you receive monthly granularity (YYYYMM), derive quarter via:
- `quarter = ceil(month/3)`
- If you receive a range (start_month..end_month), allocate by:
- **equal-month weighting** unless you have usage evidence to weight by plays/units.
- For retro adjustments that specify a multi-year range (common in settlements):
- store the adjustment as an `AccountTransaction` with a `covers_activity_range`,
- optionally allocate into the activity ledger if the statement supplies enough specificity.

Currency / FX rules

Use DDEX’s conceptual separation as your canonical model:

- `transaction_currency`
- `reporting_currency`
- `invoicing_currency`

…and record conversion details explicitly. [16]

Minimum FX policy metadata:

Field	Purpose
fx_source	e.g., central bank series, vendor-provided
fx_rate	numeric
fx_timestamp_rule	avg over activity period vs statement-end vs settlement date
fx_retranslation_rule	whether reversals use original fx

Identifier mapping, fuzzy matching, and manual overrides

Identifier mapping blueprint

- **Work mapping**: internal WorkUID ↔ ISWC ↔ society IDs ↔ publisher/admin IDs
  ISWC is standardized for globally unique musical-work identification. [17]
- **Recording mapping**: internal RecordingUID ↔ ISRC ↔ DSP/label IDs
  ISRC is a unique identifier for each recording/version. [9]
- **Party mapping**: internal PartyUID ↔ IPI ↔ ISNI ↔ society payee IDs
  IPI uniquely identifies interested parties and supports distribution/accounting. [7]
  ISNI is the ISO name identifier used across rights/metadata ecosystems. [18]

ISRC ↔ ISWC crosswalk for master vs publishing reconciliation

A “complete” royalty app must reconcile composition revenue streams (mechanical/performance/sync) with master revenue streams (recording exploitation). Your KB should store **many-to-many** edges:

- One ISWC → many ISRCs (covers, remixes, live versions)
- One ISRC → potentially multiple ISWCs (medleys, interpolations, samples—depending on reporting granularity)

A key authoritative anchor: the US Copyright Office’s rules for the MLC database explicitly include both work identifiers (ISWC) and sound recording identifiers (ISRC) and require the database to identify sound recordings in which works are embodied. [14]

Fuzzy matching rules

You need deterministic-first logic:

1) **Exact ID matches** (ISWC/ISRC/provider IDs).
2) **Exact normalized title + exact normalized party set**.
3) **Fuzzy title similarity + partial party overlap**.

Recommended scoring:

- Title similarity (token-set ratio): 0–60 points
- Party match overlap (Jaccard): 0–30 points
- Duration/version match (if present): 0–10 points

- Thresholds:
- ≥85: auto-match
- 70–84: “review required”
- <70: do not match automatically

Manual override workflow

Overrides must be first-class records with audit trails:

- `override_uid`
- `created_by`
- `created_at`
- `reason_code` (e.g., duplicate work IDs, provider mislabel, dispute)
- `effective_start` / `effective_end`
- `evidence_refs` (links to statement lines/pages/files)
- `supersedes_mapping_uid`

This is essential because disputes and audits are not theoretical; formal audit programs exist in the ecosystem, and your app should anticipate similar governance requirements. [19]

Classification and reconciliation modules with worked tables

Line-item classification taxonomy

Your KB should classify each `StatementLine` along three axes:

1) **Right category** (high-level) - Mechanical - Performance - Synchronization (sync) - Print/Lyric - Neighboring rights / sound recording performance (master-side statutory regimes) - Other

2) **Channel / product** - Streaming, download, physical, UGC, broadcast, live, AV, social short-form

3) **Accounting flag** - current-period earned - retro adjustment (true-up) - withholding tax - reserve movement - recoupment - non-royalty cost chargeback

For performance income, PROs themselves describe how royalties are divided and computed. For example, **Broadcast Music, Inc.** [20] frames distributions as a 200% unit split into writer and publisher shares (typically 100% each). [21]
Similarly, **American Society of Composers, Authors and Publishers** [22] publishes a high-level calculation formula: credits × share × credit value. [23]

For master-side statutory digital performance contexts, **SoundExchange, Inc.** [24] describes Reports of Use requirements and the need for detailed sound recording performance reporting (the ISRC prominence is consistent with this ecosystem). [25]

Statement parser expectations (examples)

MLC parser - Support both statement layouts: - Royalty Detail (recording-level, aligned to DSP usage reports) - Work Summary (work-level aggregation) [26] - Persist “as-reported” calculation components; streaming mechanical rates are formula-driven under CRB regulations (percent of revenue vs TCC constructs). [27] - Preserve adjustment provenance (original reporting period). [15]

PRO parser - Expect pool-based / credit/weighting logic rather than fixed per-stream rates; store usage evidence references when present. [28]

SoundExchange parser - Model Report of Use ingestion as recording-first (ISRC-first) and audience measurement fields. [29]

Reconciliation algorithms

A complete app needs **three reconciliation families**.

Control totals (statement-level)

Core identities to reconcile:

- `Σ(gross_line_amounts) = statement_gross_payable`
- `Σ(withholding_tax_lines) = statement_withholding_tax`
- `Σ(net_line_amounts) = statement_net_payable`

Balance roll-forward (account-level)

- `closing_balance = opening_balance + net_payable_royalties + adjustments - payments - advances +/- other`

Share integrity checks

At any aggregation granularity (work, right type, territory, period):

- Sum of payable shares should equal 1.0 (or 100%) for the payable basis you are modeling.
- Any deviations route to suspense/unmatched/dispute.

Worked reconciliation tables (Computed from uploaded statements)

Your uploaded dataset includes statement PDFs plus detailed CSV exports for the same periods. This is a best-case ingestion outcome: you can reconcile summary-to-detail exactly.

Below is the computed match between detailed CSV totals and statement summary totals for each statement period (gross, withholding, net). Values are USD.

Statement period	Gross payable (sum of lines)	Withholding tax (sum of lines)	Net payable (sum of lines)
2022-07-01 to 2022-12-31	17,092.67	-21.62	17,071.05
2023-01-01 to 2023-06-30	49,161.99	-33.43	49,128.56
2023-07-01 to 2023-12-31	47,970.07	-89.86	47,880.21
2024-01-01 to 2024-06-30	46,214.67	-76.91	46,137.77
2024-07-01 to 2024-12-31	41,057.85	-180.63	40,877.21
2025-01-01 to 2025-06-30	56,886.26	-62.66	56,823.60

This is your reference "gold standard" control check for the app: the parser should reproduce these totals exactly (within rounding).

Classification example (Computed from uploaded statements)

Income-type mix for 2024-07-01 to 2024-12-31 (net, after withholding):

Income type	Net payable
Streaming Mechanical	23,938.26
Streaming Performance	8,688.06
Digital Synch	3,126.45
Live Performance	827.49
Download Mechanical	807.85
Digital Performance	783.93
Digital Lyrics	470.73
Performance	409.97
Radio Performance	404.82
Digital Mechanical	349.20

This illustrates why your KB should store both **raw provider "income types"** and a **canonical category** mapping layer.

Time-series analytics, decay modeling, and valuation modules

Analysis pipeline flowchart

```
flowchart TD
  A[Canonical StatementLine Ledger] --> B[Build Time Series\nper-work, per-right, per-territory]
  B --> C[Outlier + Retro Logic\nseparate earned vs cash]
  C --> D[Decay Metrics\nCAGR, exp fit, k, half-life, R²]
  D --> E[Portfolio Analytics\nconcentration, segmentation]
  E --> F[Forecast Scenarios\nbase/downside/upside]
  F --> G[Valuation Engine\nDCF + multipliers + sensitivity]
  G --> H[Reports + APIs\nbuyer pack + internal QA]
```

Per-song decay metrics

Your KB should implement at least two decay modes:

1) **Peak-anchored exponential decay** (post-peak tail)

$$y_t = y_0 e^{-kt}$$

Half-life:

$$t_{1/2} = \frac{\ln 2}{k}$$

2) **CAGR** over comparable intervals (use sparingly where series is stable)

$$\mathrm{CAGR} = \left(\frac{y_T}{y_0}\right)^{\frac{1}{\Delta t}} - 1$$

You should compute fit quality (log-space R^2) because many works do not follow a clean exponential tail (multiple re-activations, sync shocks, platform changes).

Worked decay example (Computed from uploaded statements)

Example work ("PUSSY & MILLIONS") built as net payable aggregated by activity quarter (distribution end quarter), with a peak-anchored exponential fit.

Fit parameters: - Peak quarter: 2022Q4
- $k \approx 0.345$ per quarter
- Half-life $t_{1/2} \approx 2.01$ quarters (~6.0 months)
- log-space $R^2 \approx 0.86$

Post-peak observed vs fitted (net):

Quarter	Observed	Fitted
2022Q4	18,179.79	18,179.79
2023Q1	10,322.44	12,874.48
2023Q2	14,958.47	9,117.38
2023Q3	6,638.04	6,456.70
2023Q4	4,025.31	4,572.48
2024Q1	8,387.58	3,238.11
2024Q2	2,372.96	2,293.15
2024Q3	1,961.48	1,623.95
2024Q4	1,219.43	1,150.04
2025Q1	370.15	814.43

```
xychart-beta
  title "Decay overlay (Computed from uploaded statements): PUSSY & MILLIONS"
  x-axis
["2022Q4","2023Q1","2023Q2","2023Q3","2023Q4","2024Q1","2024Q2","2024Q3","2024Q4","2025Q1"]
  y-axis "Net payable (USD)" 0 --> 20000
  line
"Observed" [18179.79,10322.44,14958.47,6638.04,4025.31,8387.58,2372.96,1961.48,1219.43,370.15]
  line "Exp
fit"  [18179.79,12874.48,9117.38,6456.70,4572.48,3238.11,2293.15,1623.95,1150.04,814.43]
```

Portfolio decay and concentration (Computed from uploaded statements)

Portfolio series (net payable summed across all works) and post-peak exponential fit:

- Peak quarter: 2023Q2
- Peak net: 36,398.27
- $k \approx 0.197$ per quarter
- Half-life ≈ 3.51 quarters (~10.5 months)
- log-space R^2 ≈ 0.60

```
xychart-beta
  title "Portfolio observed vs exp fit (Computed from uploaded statements)"
  x-axis
["2023Q2","2023Q3","2023Q4","2024Q1","2024Q2","2024Q3","2024Q4","2025Q1","2025Q2"]
  y-axis "Net payable (USD)" 0 --> 40000
  line
"Observed" [36398.27,21011.67,21756.57,34733.87,20280.35,13191.05,14677.20,16369.24,2948.29]
```

```
  line "Exp
fit"  [36398.27,29876.00,24522.47,20128.25,16521.43,13560.93,11130.92,9136.35,7499.19]
```

Concentration metrics your KB should compute per period: - Top-1 share - Top-5 share - HHI (sum of squared shares)

Example (selected quarters; computed from uploaded statements):

Quarter	Total net	Top-1 share	Top-5 share	HHI
2023Q2	36,398.27	0.4110	0.8120	0.2182
2024Q1	34,733.87	0.2415	0.7143	0.1342
2025Q1	16,369.24	0.4416	0.7599	0.2307

Sensitivity scenarios

Your KB should ship default scenario logic:

- **Exclude non-recurring categories** (e.g., one-time sync settlements) from normalized run-rate.
- **Earned vs cash**:
- "Earned" series: line-item royalties allocated by activity period.
- "Cash" series: add `AccountTransaction` flows (payments, advances, retro adjustments) by posting date.
- **Territory normalization sensitivity**: show results with and without territory canonicalization to quantify how much messy geo labels distort reporting.

Treatment of advances, recoupment, reserves, and retro adjustments

Store these as separate, explicit accounting objects:

- `Advance`: cash paid upfront; often recouped from future royalties.
- `Recoupment`: reductions of payable/cash until advance is recovered.
- `Reserve`: withheld portion (more common in physical/return contexts, but still appears in certain contracts).
- `RetroAdjustment`: true-ups for prior periods; should retain original period coverage when known (benchmark principle reflected in MLC adjustment reporting rules). [15]

Valuation module

Your KB should support two valuation "views":

1) **Income-multiple view**
- Inputs: normalized annual net, growth/decay band, concentration risk premium/discount. - Outputs: valuation band.

2) **DCF view**
- Inputs: forecast by channel, decay k (or segmented ks), discount rate, terminal value method, scenario matrix. - Outputs: NPV per scenario and sensitivity tables.

Also include tax/amortization notes for buyer-side modeling. In the US, acquired “section 197 intangibles” are generally amortized over 15 years; the IRS describes this directly, and 26 U.S.C. §197 is the statutory anchor. [30]

Example DCF template (structure your KB should encode):

Year	Forecast net	Growth/decay	Discount factor	PV
1	N_1	g_1	$1/(1+r)^1$	PV_1
…	…	…	…	…
Terminal	N_T × multiple	—	1/(1+r)^T	PV_T

Terminal value should be configurable (exit multiple or Gordon growth). Sensitivities must include at least: - discount rate, - decay/growth, - removal/addback of one-time events, - concentration haircut.

Delivery-ready machine pack (JSON and YAML)

This section is designed to be pasted directly into your app’s KB store. It includes: schema stubs, rule dictionaries, reconciliation checks, analytics formulas, and example records.

JSON package

```
{
  "kb_version": "1.0.0",
  "kb_name": "rhythm_catalog_kb",
  "principles": {
    "immutability": "Raw ingests are immutable; all transforms are
reproducible.",
    "provenance": "Every normalized field must retain source_file/source_row or
source_page reference.",
    "three_time_axes": ["activity_period", "statement_period",
"cash_posting_date"],
    "earned_vs_cash": "Earned series is activity-based; cash series includes
account transactions."
  },
  "schemas": {
    "Catalog": {
      "required": ["uid", "name", "base_currency", "timezone"],
      "fields": {
        "uid": "string",
        "name": "string",
```

```
        "base_currency": "ISO4217",
        "timezone": "IANA",
        "catalog_scope": {
          "includes": ["publishing", "master"],
          "territory_scope": ["WORLD", "US", "GB"],
          "effective_start": "date",
          "effective_end": "date|null"
        }
      }
    },
    "Party": {
      "required": ["uid", "name"],
      "fields": {
        "uid": "string",
        "name": "string",
        "party_type": "enum:rightsholder|publisher|admin|label|pro|cmo|dsp|
payee|vendor",
        "external_ids": [
          { "scheme": "enum:IPI|ISNI|DDEX_DPID|PRO_ID|VENDOR_ID", "value":
"string" }
        ]
      }
    },
    "Work": {
      "required": ["uid", "title"],
      "fields": {
        "uid": "string",
        "title": "string",
        "iswc": "string|null",
        "alt_titles": ["string"],
        "writers": [
          {
            "party_uid": "string",
            "ipi": "string|null",
            "share_type": "enum:manuscript|collection|payable",
            "share_pct": "number"
          }
        ],
        "publishers_admins": [
          {
            "party_uid": "string",
            "ipi": "string|null",
            "role": "enum:publisher|administrator|subpublisher",
            "share_pct": "number"
          }
        ]
      }
    },
```

```
    "Recording": {
      "required": ["uid", "title"],
      "fields": {
        "uid": "string",
        "title": "string",
        "isrc": "string|null",
        "upc": "string|null",
        "artists": ["string"],
        "linked_works": [
          { "work_uid": "string", "allocation_rule": "enum:equal|usage_weighted|manual", "weight": "number|null" }
        ]
      }
    },
    "Statement": {
      "required": ["uid", "issuer", "statement_period", "currency", "source_type"],
      "fields": {
        "uid": "string",
        "issuer": "string",
        "source_type": "enum:publisher_admin|pro|mlc|soundexchange|label_distributor|other",
        "statement_period": { "start": "date", "end": "date" },
        "currency": "ISO4217",
        "settlement_lag_days": "number|null",
        "totals": {
          "gross_payable": "number|null",
          "withholding_tax": "number|null",
          "net_payable": "number|null"
        },
        "raw_refs": { "files": ["string"], "notes": "string|null" }
      }
    },
    "StatementLine": {
      "required": ["uid", "statement_uid", "amount", "currency", "activity_period"],
      "fields": {
        "uid": "string",
        "statement_uid": "string",
        "work_uid": "string|null",
        "recording_uid": "string|null",
        "activity_period": { "start": "date|null", "end": "date|null", "year_month": "string|null" },
        "territory_raw": "string|null",
        "territory_iso2": "string|null",
        "source_provider": "string|null",
        "exploitation_source": "string|null",
        "units": "number|null",
```

```
        "amount": "number",
        "currency": "ISO4217",
        "withholding_tax": "number|null",
        "net_amount": "number|null",
        "rate_pct": "number|null",
        "raw_income_type": "string|null",
        "canonical_right_category": "enum:mechanical|performance|sync|
print_lyrics|neighboring_rights|other",
        "canonical_channel": "enum:streaming|download|broadcast|live|av|ugc|
social|physical|other",
        "accounting_flags": {
          "is_retro_adjustment": "boolean",
          "is_withholding": "boolean",
          "is_reserve_movement": "boolean",
          "is_recoupment": "boolean",
          "is_nonroyalty_charge": "boolean"
        },
        "provenance": { "source_file": "string", "source_row": "number|null" }
      }
    },
    "AccountTransaction": {
      "required": ["uid", "statement_uid", "txn_type", "amount", "currency",
"posting_date"],
      "fields": {
        "uid": "string",
        "statement_uid": "string",
        "txn_type": "enum:payment|advance|recoupment|chargeback|
retro_adjustment|reserve",
        "posting_date": "date",
        "covers_activity_range": { "start": "date|null", "end": "date|null" },
        "amount": "number",
        "currency": "ISO4217",
        "memo": "string|null"
      }
    }
  },
  "rules": {
    "territory_normalization": {
      "method": "dictionary_then_fuzzy",
      "dictionary": {
        "US": "US",
        "United States": "US",
        "UNITED STATES": "US",
        "GB": "GB",
        "United Kingdom": "GB",
        "UNITED KINGDOM": "GB"
      }
    },
```

```
    "currency_fx": {
      "base_currency_policy": "catalog.base_currency",
      "fx_fields": ["transaction_currency", "reporting_currency",
"invoicing_currency"],
      "fx_timestamp_rule_default": "average_over_activity_period"
    },
    "classification": {
      "right_category_rules": [
        { "if_raw_income_type_contains": ["Mechanical"], "set_right_category":
"mechanical" },
        { "if_raw_income_type_contains": ["Performance"], "set_right_category":
"performance" },
        { "if_raw_income_type_contains": ["Synch", "Synchronisation"],
"set_right_category": "sync" },
        { "if_raw_income_type_contains": ["Print", "Lyrics"],
"set_right_category": "print_lyrics" }
      ],
      "channel_rules": [
        { "if_raw_income_type_contains": ["Streaming"], "set_channel":
"streaming" },
        { "if_raw_income_type_contains": ["Download"], "set_channel":
"download" },
        { "if_raw_income_type_contains": ["Radio", "TV", "Film"],
"set_channel": "broadcast" },
        { "if_raw_income_type_contains": ["Live"], "set_channel": "live" }
      ]
    },
    "fuzzy_matching": {
      "title_normalization": "upper_trim_strip_punct_collapse_ws",
      "party_normalization": "upper_trim_strip_punct",
      "auto_match_threshold": 85,
      "review_threshold": 70,
      "scoring_weights": { "title": 60, "parties": 30, "duration_or_version":
10 }
    },
    "reconciliation_checks": {
      "control_totals": [
        "sum(StatementLine.amount) == Statement.totals.gross_payable",
        "sum(StatementLine.withholding_tax) ==
Statement.totals.withholding_tax",
        "sum(StatementLine.net_amount) == Statement.totals.net_payable"
      ],
      "balance_rollforward": [
        "closing = opening + net_payable_royalties + adjustments - payments -
advances"
      ]
    },
    "analytics_formulas": {
```

```
      "cagr": "((yT / y0) ** (1/dt_years)) - 1",
      "exp_decay_k": "k = - sum(t * ln(y/y0)) / sum(t^2)",
      "half_life": "t_half = ln(2)/k",
      "log_r2": "1 - SSE(log(y), log(yhat))/SST(log(y))",
      "concentration_top1": "max(work_amount)/sum(work_amount)",
      "hhi": "sum((work_amount/total)^2)"
    }
  },
  "examples_computed_from_uploaded_statements": {
    "statement_totals_by_period_usd": [
      { "period": "2022-07-01/2022-12-31", "gross": 17092.67, "withholding":
-21.62, "net": 17071.05 },
      { "period": "2023-01-01/2023-06-30", "gross": 49161.99, "withholding":
-33.43, "net": 49128.56 },
      { "period": "2023-07-01/2023-12-31", "gross": 47970.07, "withholding":
-89.86, "net": 47880.21 },
      { "period": "2024-01-01/2024-06-30", "gross": 46214.67, "withholding":
-76.91, "net": 46137.77 },
      { "period": "2024-07-01/2024-12-31", "gross": 41057.85, "withholding":
-180.63, "net": 40877.21 },
      { "period": "2025-01-01/2025-06-30", "gross": 56886.26, "withholding":
-62.66, "net": 56823.60 }
    ],
    "decay_example_work": {
      "work_uid": "work:WW01577118500",
      "title": "PUSSY & MILLIONS",
      "peak_quarter": "2022Q4",
      "k_per_quarter": 0.345,
      "half_life_quarters": 2.01,
      "r2_log": 0.86
    }
  }
}
```

YAML package

```
kb_version: "1.0.0"
kb_name: rhythm_catalog_kb

principles:
  immutability: "Raw ingests are immutable; all transforms are reproducible."
  provenance: "Every normalized field must retain source references."
  three_time_axes: ["activity_period", "statement_period", "cash_posting_date"]
  earned_vs_cash: "Earned series is activity-based; cash series includes
account transactions."
```

```yaml
schemas:
  Catalog:
    required: ["uid", "name", "base_currency", "timezone"]
    fields:
      uid: string
      name: string
      base_currency: ISO4217
      timezone: IANA
      catalog_scope:
        includes: [publishing, master]
        territory_scope: [WORLD, US, GB]
        effective_start: date
        effective_end: date|null

  Party:
    required: ["uid", "name"]
    fields:
      uid: string
      name: string
      party_type: "rightsholder|publisher|admin|label|pro|cmo|dsp|payee|vendor"
      external_ids:
        - scheme: "IPI|ISNI|DDEX_DPID|PRO_ID|VENDOR_ID"
          value: string

  Work:
    required: ["uid", "title"]
    fields:
      uid: string
      title: string
      iswc: string|null
      alt_titles: [string]
      writers:
        - party_uid: string
          ipi: string|null
          share_type: "manuscript|collection|payable"
          share_pct: number
      publishers_admins:
        - party_uid: string
          ipi: string|null
          role: "publisher|administrator|subpublisher"
          share_pct: number

  Recording:
    required: ["uid", "title"]
    fields:
      uid: string
      title: string
      isrc: string|null
```

```
      upc: string|null
      artists: [string]
      linked_works:
        - work_uid: string
          allocation_rule: "equal|usage_weighted|manual"
          weight: number|null

  Statement:
    required: ["uid", "issuer", "statement_period", "currency", "source_type"]
    fields:
      uid: string
      issuer: string
      source_type: "publisher_admin|pro|mlc|soundexchange|label_distributor|
other"
      statement_period:
        start: date
        end: date
      currency: ISO4217
      settlement_lag_days: number|null
      totals:
        gross_payable: number|null
        withholding_tax: number|null
        net_payable: number|null
      raw_refs:
        files: [string]
        notes: string|null

  StatementLine:
    required: ["uid", "statement_uid", "amount", "currency", "activity_period"]
    fields:
      uid: string
      statement_uid: string
      work_uid: string|null
      recording_uid: string|null
      activity_period:
        start: date|null
        end: date|null
        year_month: string|null
      territory_raw: string|null
      territory_iso2: string|null
      source_provider: string|null
      exploitation_source: string|null
      units: number|null
      amount: number
      currency: ISO4217
      withholding_tax: number|null
      net_amount: number|null
      rate_pct: number|null
```

```
      raw_income_type: string|null
      canonical_right_category: "mechanical|performance|sync|print_lyrics|
neighboring_rights|other"
      canonical_channel: "streaming|download|broadcast|live|av|ugc|social|
physical|other"
      accounting_flags:
        is_retro_adjustment: boolean
        is_withholding: boolean
        is_reserve_movement: boolean
        is_recoupment: boolean
        is_nonroyalty_charge: boolean
      provenance:
        source_file: string
        source_row: number|null

  AccountTransaction:
    required: ["uid", "statement_uid", "txn_type", "amount", "currency",
"posting_date"]
    fields:
      uid: string
      statement_uid: string
      txn_type: "payment|advance|recoupment|chargeback|retro_adjustment|reserve"
      posting_date: date
      covers_activity_range:
        start: date|null
        end: date|null
      amount: number
      currency: ISO4217
      memo: string|null

rules:
  territory_normalization:
    method: dictionary_then_fuzzy
    dictionary:
      US: US
      "United States": US
      "UNITED STATES": US
      GB: GB
      "United Kingdom": GB
      "UNITED KINGDOM": GB

  currency_fx:
    base_currency_policy: catalog.base_currency
    fx_fields: [transaction_currency, reporting_currency, invoicing_currency]
    fx_timestamp_rule_default: average_over_activity_period

  classification:
    right_category_rules:
```

```yaml
      - if_raw_income_type_contains: ["Mechanical"]
        set_right_category: mechanical
      - if_raw_income_type_contains: ["Performance"]
        set_right_category: performance
      - if_raw_income_type_contains: ["Synch", "Synchronisation"]
        set_right_category: sync
      - if_raw_income_type_contains: ["Print", "Lyrics"]
        set_right_category: print_lyrics
    channel_rules:
      - if_raw_income_type_contains: ["Streaming"]
        set_channel: streaming
      - if_raw_income_type_contains: ["Download"]
        set_channel: download
      - if_raw_income_type_contains: ["Radio", "TV", "Film"]
        set_channel: broadcast
      - if_raw_income_type_contains: ["Live"]
        set_channel: live

  fuzzy_matching:
    title_normalization: upper_trim_strip_punct_collapse_ws
    party_normalization: upper_trim_strip_punct
    auto_match_threshold: 85
    review_threshold: 70
    scoring_weights:
      title: 60
      parties: 30
      duration_or_version: 10

  reconciliation_checks:
    control_totals:
      - "sum(StatementLine.amount) == Statement.totals.gross_payable"
      - "sum(StatementLine.withholding_tax) == Statement.totals.withholding_tax"
      - "sum(StatementLine.net_amount) == Statement.totals.net_payable"
    balance_rollforward:
      - "closing = opening + net_payable_royalties + adjustments - payments -
advances"

  analytics_formulas:
    cagr: "((yT / y0) ** (1/dt_years)) - 1"
    exp_decay_k: "k = - sum(t * ln(y/y0)) / sum(t^2)"
    half_life: "t_half = ln(2)/k"
    log_r2: "1 - SSE(log(y), log(yhat))/SST(log(y))"
    concentration_top1: "max(work_amount)/sum(work_amount)"
    hhi: "sum((work_amount/total)^2)"

examples_computed_from_uploaded_statements:
  statement_totals_by_period_usd:
    - period: "2022-07-01/2022-12-31"
```

```
      gross: 17092.67
      withholding: -21.62
      net: 17071.05
    - period: "2023-01-01/2023-06-30"
      gross: 49161.99
      withholding: -33.43
      net: 49128.56
    - period: "2023-07-01/2023-12-31"
      gross: 47970.07
      withholding: -89.86
      net: 47880.21
    - period: "2024-01-01/2024-06-30"
      gross: 46214.67
      withholding: -76.91
      net: 46137.77
    - period: "2024-07-01/2024-12-31"
      gross: 41057.85
      withholding: -180.63
      net: 40877.21
    - period: "2025-01-01/2025-06-30"
      gross: 56886.26
      withholding: -62.66
      net: 56823.60

decay_example_work:
  work_uid: "work:WW01577118500"
  title: "PUSSY & MILLIONS"
  peak_quarter: "2022Q4"
  k_per_quarter: 0.345
  half_life_quarters: 2.01
  r2_log: 0.86
```

Source hierarchy your KB should prioritize

- **Regulations / primary law**: CFR/USC (e.g., reporting requirements, database ownership, rate calculation frameworks). [31]
- **Official organizations' standards & documentation**:
- CISAC identifiers and formats (ISWC, IPI, CWR). [32]
- DDEX standards (DSR, MWDR, currency guidance). [33]
- IFPI ISRC handbook/standard pages. [34]
- **Collection entities' published rules**:
- PRO distribution logic (BMI 200% split; ASCAP formula). [35]
- SoundExchange reporting requirements and statutory-license framing. [36]
- **Tax authorities**:
- IRS guidance and statutory anchors for amortization of acquired intangibles. [30]

This KB package is designed so Rhythm can ingest real statements, reconcile them to summary totals, normalize IDs/territories/currencies, compute decay and concentration, and produce valuation scenarios with explicit audit trails—without relying on hidden assumptions.

---

1 3 8 17 32 International Identifiers
https://www.cisac.org/services/information-services/international-identifiers?utm_source=chatgpt.com

2 13 15 22 24 31 Code of Federal Regulations 37CFR210.29
https://www.copyright.gov/title37/210/37cfr210-29.html?utm_source=chatgpt.com

4 ISRC Standard - IFPI
https://isrc.ifpi.org/isrc-standard?utm_source=chatgpt.com

5 18 20 What is ISNI?
https://isni.org/page/what-is-isni?utm_source=chatgpt.com

6 33 Digital Sales Reporting Message Suite
https://ddex.net/standards/digital-sales-reporting-message-suite/?utm_source=chatgpt.com

7 IPI
https://www.cisac.org/services/information-services/ipi?utm_source=chatgpt.com

9 34 International Standard Recording Code (ISRC) Handbook
https://www.ifpi.org/wp-content/uploads/2021/02/ISRC_Handbook.pdf?utm_source=chatgpt.com

10 Musical Works Data and Rights Communication Standards
https://ddex.net/standards/musical-works-data-and-rights-communication/?utm_source=chatgpt.com

11 16 Currency conversion in DSR and CDM
https://kb.ddex.net/implementing-each-standard/best-practices-for-all-ddex-standards/general-guidance-on-messages/currency-conversion-in-dsr-and-cdm?utm_source=chatgpt.com

12 19 DSP Audits
https://www.themlc.com/dsp-audits?utm_source=chatgpt.com

14 Code of Federal Regulations 37CFR210.31 | U.S. Copyright Office
https://www.copyright.gov/title37/210/37cfr210-31.html?utm_source=chatgpt.com

21 35 General Royalty Information
https://www.bmi.com/creators/royalty/general_information?utm_source=chatgpt.com

23 28 How ASCAP Calculates Royalties
https://www.ascap.com/help/royalties-and-payment/payment/royalties?utm_source=chatgpt.com

25 29 36 Reporting Requirements - SoundExchangeSoundExchange
https://www.soundexchange.com/service-provider/reporting-requirements/?utm_source=chatgpt.com

26 What do the different statement types mean?
https://help.themlc.com/en/support/what-do-the-different-statement-layouts-mean?utm_source=chatgpt.com

27 37 CFR § 385.21 Royalty rates and calculations
https://ecfr.io/Title-37/Section-385.21?utm_source=chatgpt.com

30 Intangibles | Internal Revenue Service
https://www.irs.gov/businesses/small-businesses-self-employed/intangibles?utm_source=chatgpt.com